

Auto Ticket Classification using Flow Designer

3rd – BSc., Artificial Intelligence & Machine Learning

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Problem Statement

The school IT helpdesk receives multiple incident requests daily from students and teachers, such as Wi-Fi issues, projector failures, password problems, and slow computers.

Currently, IT staff manually reviews each request and assigns a category, which is time-consuming and inefficient.

This project automates ticket classification using ServiceNow Flow Designer by analyzing keywords in the incident's short description and description fields.

Objective

- Automatically classify incidents at the time of creation.
- Reduce manual effort for IT agents.
- Improve ticket routing efficiency.
- Implement a no-code, easily maintainable solution.

PHASE 1: Requirement Analysis & Planning

The school IT helpdesk receives multiple support requests daily from students and teachers related to common issues such as network connectivity, hardware failures, account access problems, and system performance issues.

Currently, IT staff manually review each ticket to:

- Identify the issue type.
- Select the correct category and subcategory.

- Inform the caller about ticket creation.

This manual process is time-consuming, error-prone, and not scalable as the number of requests increases.

Activity : 1 Business Use Case.

Business Requirements

The system must:

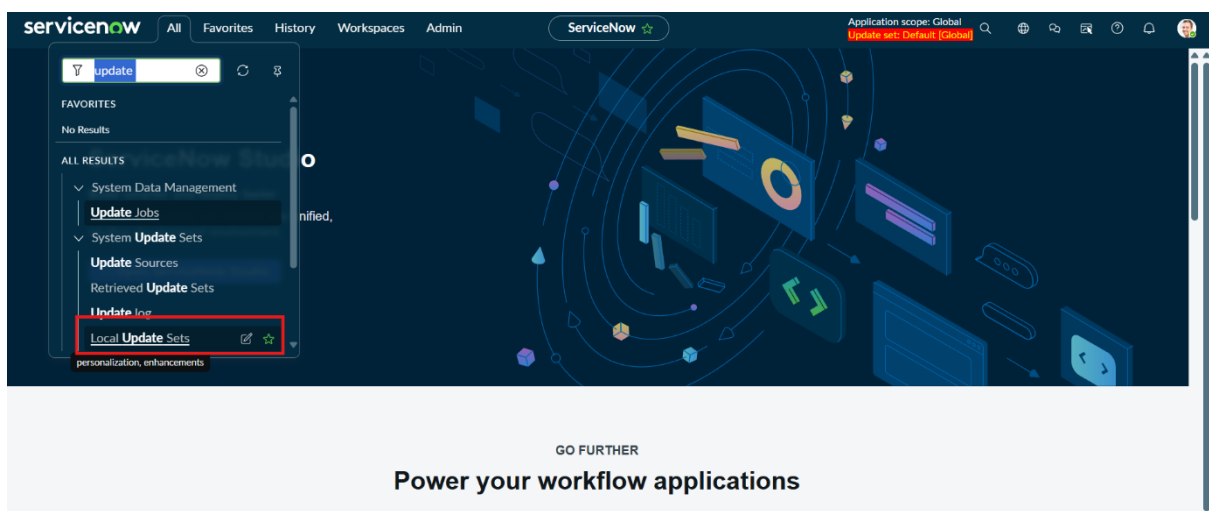
1. Automatically classify IT tickets based on the issue description.
2. Assign both Category and Subcategory without manual intervention.
3. Support dependent choice logic between Category and Subcategory.
4. Send an automated email notification to the caller upon ticket creation.
5. Store ticket information in a structured and standardized format.
6. Ensure easy maintenance and future scalability.

Activity : 2 Creation of New Update Set.

Navigation Steps :

- Go to All menus → Search for Update set.
- Under System Update Sets → Select Local Update Sets.
- Click the New button on the top right corner of the page.
- Name : **Project Update Set.**
- State : **In progress.**
- Application : **Global.**
- Then **Save the Form.**
- Under Related links Click on the **Make This My Current Set.**
- Then the Created Update Set will appear as Current.

Activity : 3 Navigation steps in Picture View.



Update Sets							
Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
Search	Search	Search	Search	Search	Search	Search	Search
Default	Global	In progress		2026-01-06 03:50:56	system	(empty)	(empty)

Related Links

[Merge Update Sets](#)

Update Set
New record

* Name Application

State

Parent

Release date

Description

Update Set
Project Update Set

* Name Application

State

Parent

Release date

Created

Created by

Merged to

Install date

Installed from

Description

Related Links

[Make This My Current Set](#)

[Merge With Another Update Set](#)

[Scan Update Set](#)

[\(SN UHIS\) Versions \(0\)](#)

Customer Updates Update Set Logs Child Update Sets Install History

Update set = Project Update Set

Created	Type	View	Target name	Updated by	Remote update set	Action
---------	------	------	-------------	------------	-------------------	--------

servicenow All Favorites History Workspaces Admin ServiceNow

Application scope: Global
Update set: Project Update Set (Global)

Application scope: Global

Update set: Project Update Set [...]

Update set: Project Update Set (Global)

ServiceNow Studio

Build smarter and deploy faster

Access all the tools you need in one unified, intuitive development environment.

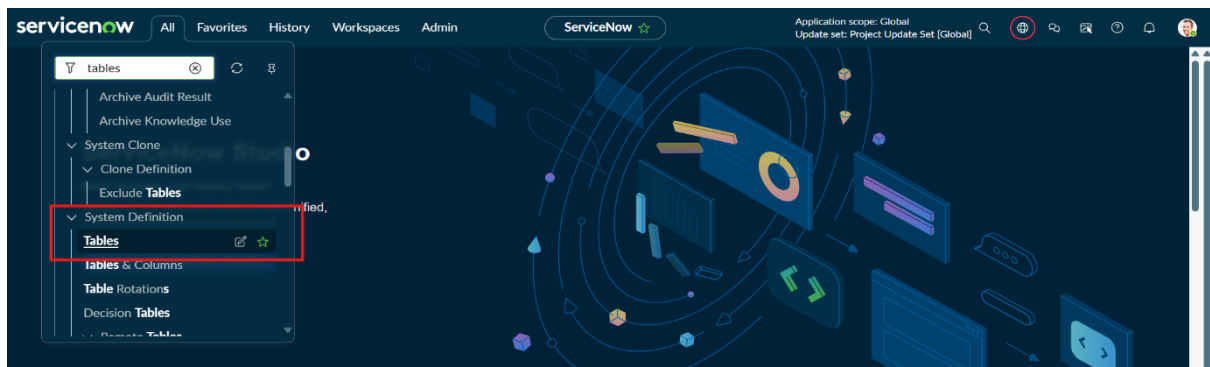
[Open ServiceNow Studio](#)

Phase :- 2 Backend Development & Configuration

Activity 1 : Creation of Custom Table to Store Ticket Records.

Steps to Create Custom Table

- Open All menus → Search for Tables.
- Under **System Definition** → Click **Tables**.
- On the top right corner Click **New** button.
- Label : **Incident WorkFlow**.
- Uncheck the **Create module**.
- Now **Save** the Form.
- Then Click the **Controls** Suction and Enable the Check box **true Auto Number**.
- Now again **save** the Form.



Label	Name	Extends table	Extensible	Updated
Search	Search	Search	Search	Search
Logged in User	v_user_session	(empty)	false	2007-07-13 16:15:23
System Plugin	v_plugin	(empty)	false	2007-10-09 12:10:05
Active Transactions	v_transaction	(empty)	false	2008-05-23 17:30:17
Node Log	v_logfiles	(empty)	false	2009-05-14 17:05:41
Customer Uploads	v_customer_uploads	(empty)	false	2009-05-14 18:38:13
MySQL Process List	v_mysql_proclist	(empty)	false	2012-02-25 19:47:02
MySQL Status	v_mysql_status	(empty)	false	2012-02-25 19:47:08
MySQL Variables	v_mysql_variables	(empty)	false	2012-02-25 19:47:12
IOStats	v_iostats	(empty)	false	2012-02-27 17:13:51
Cluster Nodes	v_cluster_nodes	(empty)	false	2012-02-27 23:48:26
SQL Debug	v_sql_debug	(empty)	false	2012-02-28 16:40:55
Workflow Validation Report	v_wf_validation_report	(empty)	false	2012-10-01 10:43:05
Hyper-V Resource Pool Component	cmdb_ci_hyper_v_rpool_comp	Hyper-V Object	false	2012-10-30 08:07:25
Application Menu	sys_app_application	Application File	false	2012-10-30 13:59:18
Menu Category	sys_app_category	Application File	false	2012-10-30 13:59:44
Scriptable Object	v_scriptable_object	(empty)	false	2012-11-09 15:15:23

Table Incident Workflow

Columns Controls Application Access

Extensible ☐

Live feed ☐

Use auto-numbering to define a sequential identifying code made up of a prefix, a base number and a padding value to ensure a consistent format

Auto-number ☒

Prefix INC

Number 500

Number of digits 5

Security Rules (ACLs) are required if anyone other than an administrator needs to work with this table. Creating default security rules will grant full access to this table to anyone with the user role you specify.

Create access controls ☒

* User role u_incident_workflow_user

Delete Update Delete All Records

Related Links

Form Builder

Table Form

The screenshot shows the 'Table' configuration page in ServiceNow. At the top, there's a header with 'Table' and 'New record' buttons. Below it, a yellow banner provides a recommendation about creating custom tables. A blue box explains that a table is a collection of records. The main form includes fields for 'Label' (Incident WorkFlow), 'Name' (u_incident_workflow), 'Application' (Global), 'Create module' (checkbox), and 'Extends table' (dropdown). Below the form is a 'Table Columns' section with a search bar and a table with columns: Column label, Type, Reference, Max length, Default value, and Display. At the bottom, there are 'Submit' and 'Cancel' buttons, and a 'Related Links' section.

- Then Save the Form and Under Related Links Click **Show Form**.
- Now Open the Configure On clicking the Context Menu and then Open the **Form Designer**.
- Then Create Fields as Required.

The screenshot shows the 'Incident WorkFlow' form in ServiceNow. The context menu is open, showing options like 'Save', 'Insert', 'Insert and Stay', 'Analyze Access', 'Configure', 'Export', 'View', 'Create Favorite', 'Copy URL', 'Copy sys_id', 'Show XML', 'History', and 'Reload form'. The 'Configure' option is highlighted, and its sub-menu is open, showing options like 'Form Builder', 'Form Design', 'Form Layout', 'Related Lists', 'All', 'Table', 'Security Rules', 'Business Rules', 'Client Scripts', 'UI Policies', 'Data Policies', 'UI Actions', 'Notifications', and 'Dictionary'. The 'Form Design' option is highlighted. The form fields include 'Number' (INC00519), 'Caller' (Janice Twiet), 'State' (None), 'Assigned group', and 'Assigned to'.

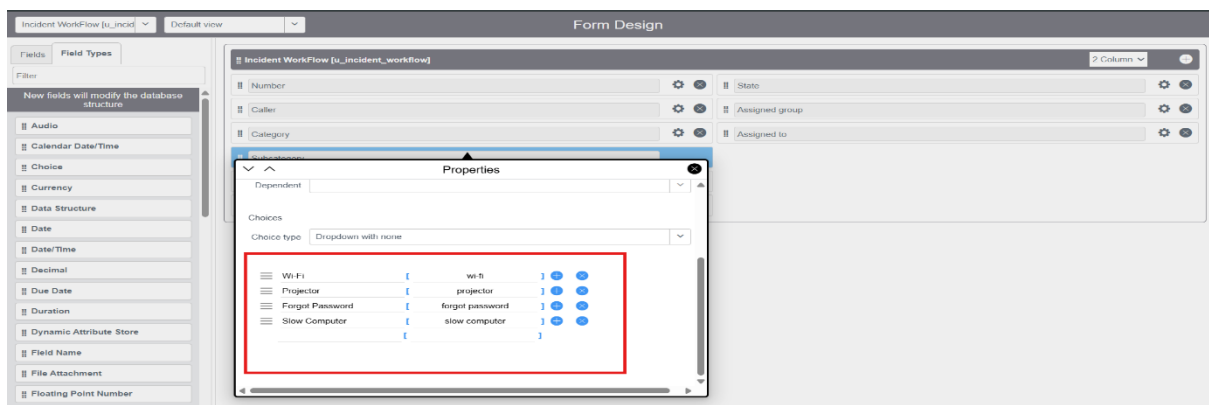
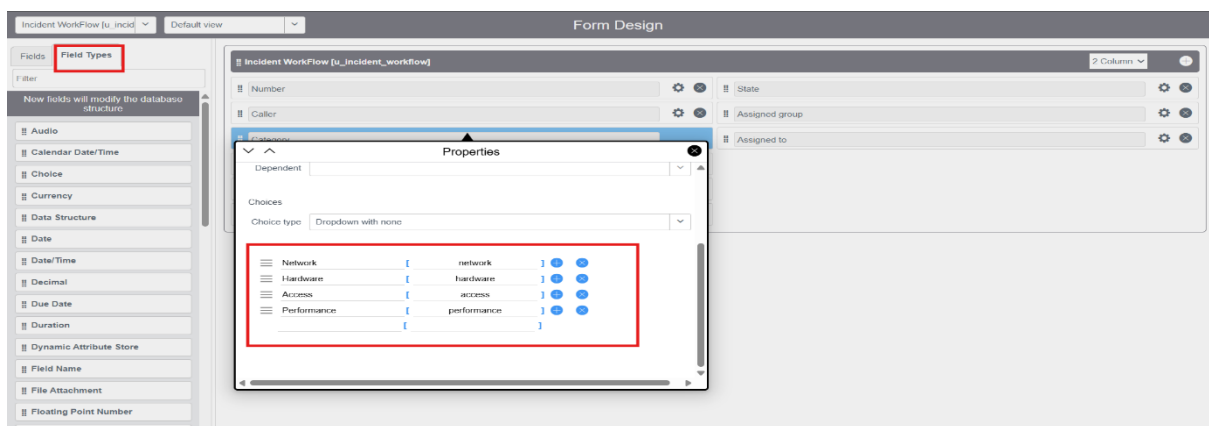
Activity 2: Field Creation and Data Type Configuration

The objective of this milestone is to create all required fields in the School IT Ticket table with correct data types, choices, and dependencies, ensuring structured and consistent data capture.

Field Label	Type	Reference/Choice
Number	Auto Number	—
Caller	Reference	sys_user
Category	Choice	[Network,Hardware,Access,Performance]
Subcategory	Choice	[Wi-Fi,Projector,Forgot Password,Slow Compter]
Short Description	String	—
Description	String	—
State	Choice	[New,In progress,On hold,Resolved,Closed]

Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

- After Creating Fields with their appropriate Type, then add the Choice from the Form design.
- To all the Choice list Fields.
- They are Category,Subcategory and State.



- Same like that, add Choice to the State field.
- Then Click the save button on the top right corner.
- Open the Form and reload it.

Activity 3 : Implementing Dependency Between Category and Subcategory Choice Fields

The objective of this milestone is to implement a **dependency relationship** between the **Category** and **Subcategory** fields so that the available Subcategory options dynamically change based on the selected Category.

Navigation Steps :

- Open the Created Table Form and Right click on the **Subcategory** Field.
- Click on the **Configure Dictionary**.
- Under **Related links** Click on the **Advanced view**,
- Then select the **Dependent Field** Section.
- Enable the checkbox **Use dependent field true**.
- Then select the Field **Category** on the **Dependent on field**.
- Then **save** the Form.

Incident Workflow
New record

Number: INC00509

State: -- None --

Assigned group: [Search]

Assigned to: [Search]

Caller: [Search]

Category: -- None --

Subcategory: [Dropdown]

Short description: [Text Area]

Details: [Text Area]

Submit

Dictionary Entry
Subcategory View: Advanced*

A dictionary entry manages how ServiceNow stores data in tables and fields (columns). For new dictionary entries, select a **Table** and the field **Type** of the new column. Also enter a column label, which becomes the field label, and the column name. If necessary, set a **Max length** for text String type fields, make the field **Mandatory** to save a record, and make the field a **Display Value** for reference fields so it appears on records that reference this table. [More Info](#)

* Table: Incident Workflow [u_incident_workflow]

* Type: Choice

* Column label: Subcategory

* Column name: u_subcategory

Application: Global

Active: ☒

Function field: ☐

Read only: ☐

Mandatory: ☐

Display: ☐

Attributes: [Text Area]

Choice List Specification: **Dependent Field** | Calculated Value | Default Value

Specify a field whose value is used to determine choice list options for this field. [More Info](#)

Use dependent field: ☒

Dependent on field: Category

- Now scroll down to the choice section.
- Then add the dependent fields for which option dependent on the category gives the category choice backend name to the subcategory choice values.
- Wi-Fi → Network.
- Projector → Hardware.
- Forgot password → Access.
- Slow computer → Performance.

Dictionary Entry Subcategory View: Advanced*

Specify a field whose value is used to determine choice list options for this field. [More Info](#)

Use dependent field ☒

Dependent on field

[Create Choice List](#) [Delete Column](#) [Update](#)

Related Links

Show Table

Run Point Scan

ISN Utilis Versions (0)

Access Controls **Choices (4)** Attributes Labels (1)

Label

Actions on selected rows... [New](#)

Label	Value	Language	Sequence	Inactive	Updated	Dependent value
☐ Forgot Password	forgot password	en	2	false	2026-02-06 21:45:02	access
☐ Wi-Fi	wi-fi	en	0	false	2026-02-06 21:44:50	network
☐ Slow Computer	slow computer	en	3	false	2026-02-06 21:45:15	performance
☐ Projector	projector	en	1	false	2026-02-06 21:45:36	hardware

[Insert a new row...](#)

- Then save the Form.
- Then it will display each choice to each one separately based on the category subcategory displays.

Incident Workflow New record

Number

State

* Caller

Assigned group

Assigned to

Category

Subcategory

* Short description

Description

[Submit](#)

Incident Workflow New record

Number

State

* Caller

Assigned group

Assigned to

Category

Subcategory

* Short description

Description

[Submit](#)

Milestone 3 : Automation using Flow Designer & Email Notification.

When a student creates a new IT support ticket, they only need to enter the Caller Name and Short Description of the issue. The student does not need to manually select the Category or Subcategory. Once the ticket is submitted, a Flow Designer automation is triggered and evaluates the Short Description for predefined keywords such as WiFi, Projector, Password, or Slow. Based on the identified keyword, the flow automatically assigns the appropriate Category and Subcategory to the ticket.

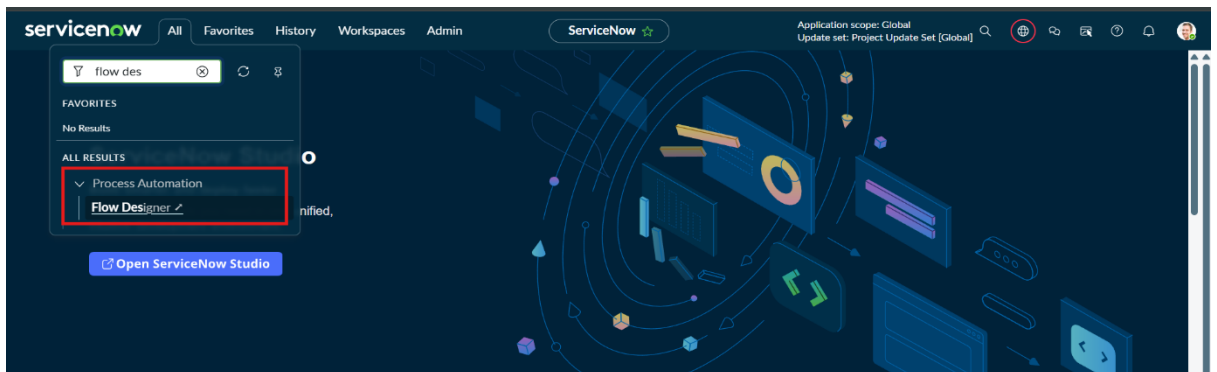
- If the short description contains the word *WiFi*, the flow assigns **Category = Network** and **Subcategory = WiFi**.
- If the description contains *Projector*, the flow assigns **Category = Hardware** and **Subcategory = Projector**.
- If the description includes *Password* or *Login*, the flow assigns **Category = Access** and **Subcategory = Forgot Password**.
- If the description mentions *Slow* or *Hanging*, the flow assigns **Category = Performance** and **Subcategory = Slow Computer**.

This automation ensures consistent classification, eliminates human error, and significantly reduces the manual effort required by IT staff.

The entire process is implemented using **Flow Designer**, making the solution easy to maintain, scalable, and completely no-code.

Activity : 1 Auto Classification using Flow design.

- Go to **All menus** → Search **Flow designer**.
- Under **Process Automation** Open the **Flow designer**.
- On the top right corner Click **New** and Select **Flow** Option.
- Name : **Auto Classify School IT Tickets**.
- Application : **Global**.
- Now Click the **Build Flow** button.



GO FURTHER Power your workflow applications

Workflow Studio

Homepage Operations Integrations

Playbooks Flows Subflows Triggers Actions Decision tables

Flows 93
Last refreshed just now.

Name	Application	Status	Active	Updated
Application Intake Request Flow (Deprecated)	Application Intake	Published	false	2026-01-31 06:37:02
Application Intake Request V2	Application Intake	Published	true	2026-01-31 06:36:56
Benchmark Recommendation Evaluator	Benchmarks Spoke	Published	true	2026-01-06 04:15:32
Business process approval flow	Global	Published	true	2020-09-27 22:06:13
Change - Cloud Infrastructure - Authorize	Global	Published	true	2020-11-11 07:08:05
Change - Conflict Detection	Global	Published	true	2026-01-06 04:06:02
Change - Emergency - Authorize	Global	Published	true	2020-10-06 05:39:49
Change - Emergency - Implement	Global	Published	true	2020-09-23 05:06:26
Change - Emergency - Review	Global	Published	true	2020-10-27 04:18:08

New

- Playbook
- Flow
- Subflow
- Trigger
- Action
- Decision table
- Help me decide

Pick up where you left off

- Major Incident Playbook - A...
Last updated: 2 d. ago by System Admi...
- Major Incident Playbook - C...
Last updated: 2 d. ago by System Admi...
- Major Incident Playbook - P...
Last updated: 2 d. ago by System Admi...

Latest updates

- System Administrator modified Major Incident Playbook - Attach Existing Communication Plans
2 d. ago
- System Administrator modified Major Incident Playbook - Create Adhoc Task
2 d. ago
- System Administrator modified Major Incident Playbook - Post Incident Report
2 d. ago

Let's get the details for your flow

Flow name *

Application *

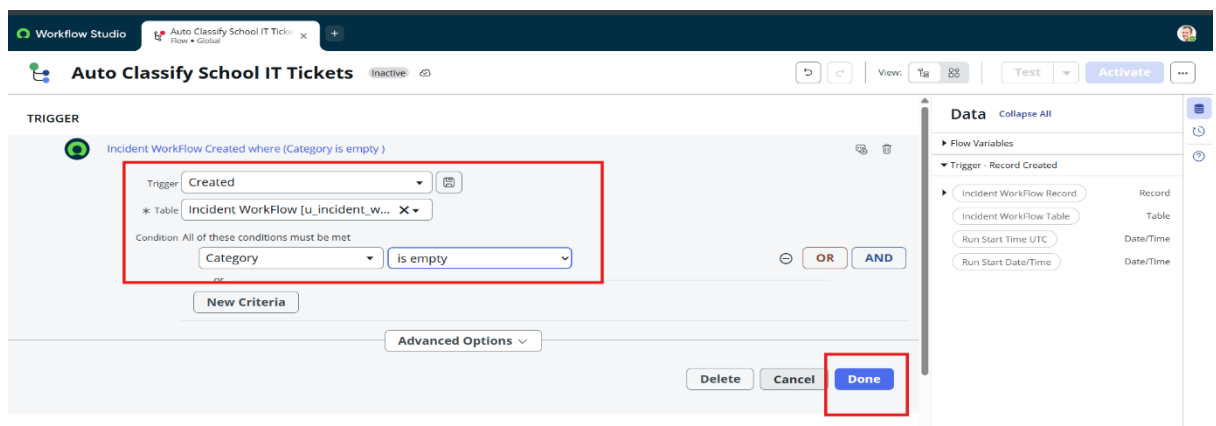
Description

[Show additional properties](#)

Cancel **Build flow**

Activity : 2 Trigger Configuration.

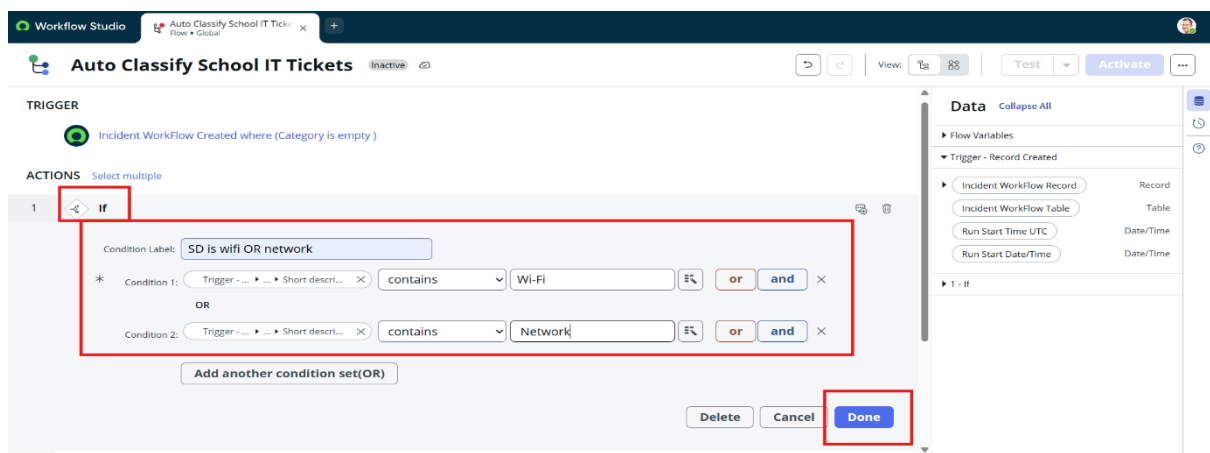
- Trigger : **Record Created.**
- Table : **Incident Workflow.**
- Condition : **Category is Empty.**
- Then Click **Done.**



- Now Under **Actions** Select **Flow Logic** Option.

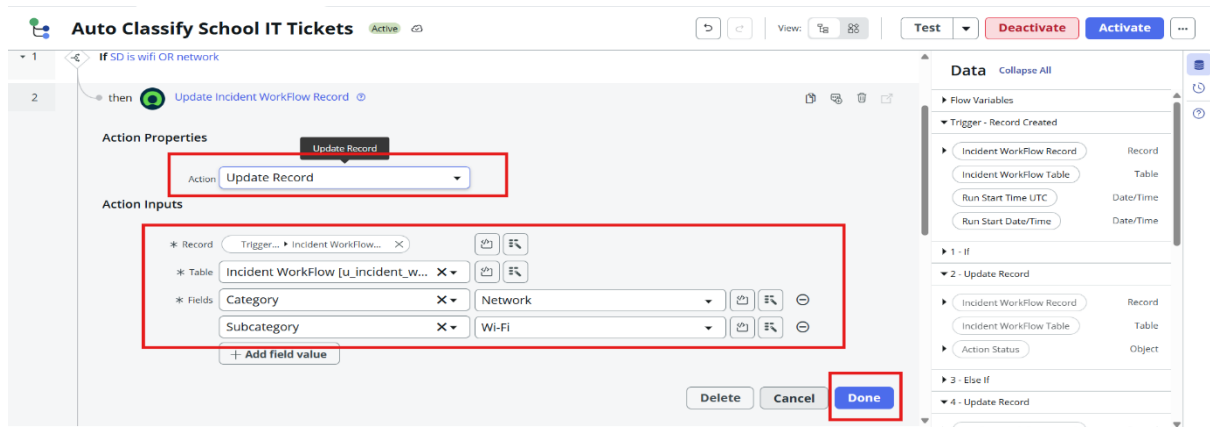
Activity : 3 Wi-Fi Issue Classification.

- Select **If** condition.
- Condition label : SD is **Wifi or Network** (Give as ur wish).
- Condition 1 : Using data pill picker Select **Short description** as Field , and operator **Contains** , Value is **Wi-Fi**.
- Condition 2 : Using data pill picker Select **Short description** as Field, and Operator is **Contains**, Value is **Network**.
- Then Click **Done** Button.



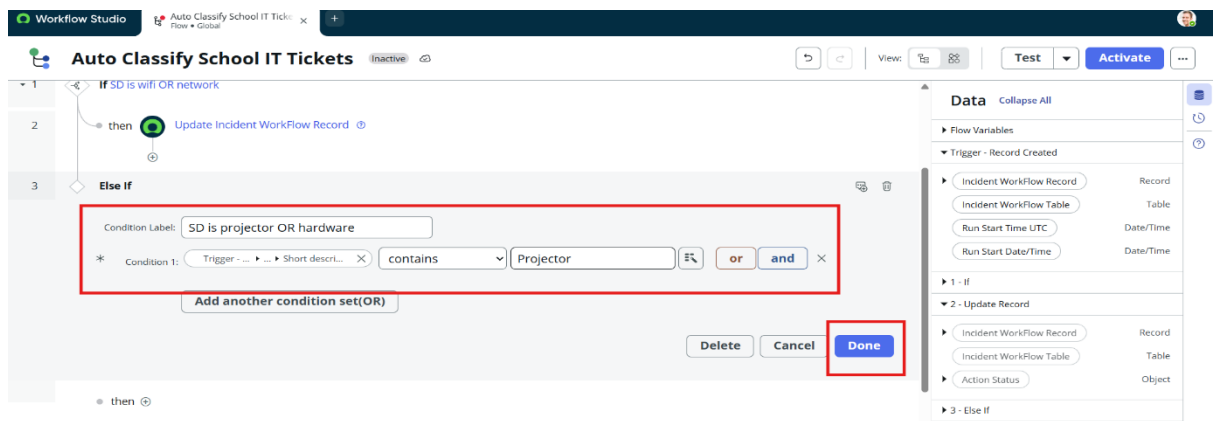
- Now Click **Action to Update record**.
- Action Inputs.
- In Record Field drag and drop data from the right side.
- Table : Auto updates as Incident Workflow.

- Fields Set **Category = Network**.
- Set **Subcategory = Wi-Fi**.
- Then Click the **Done** button.

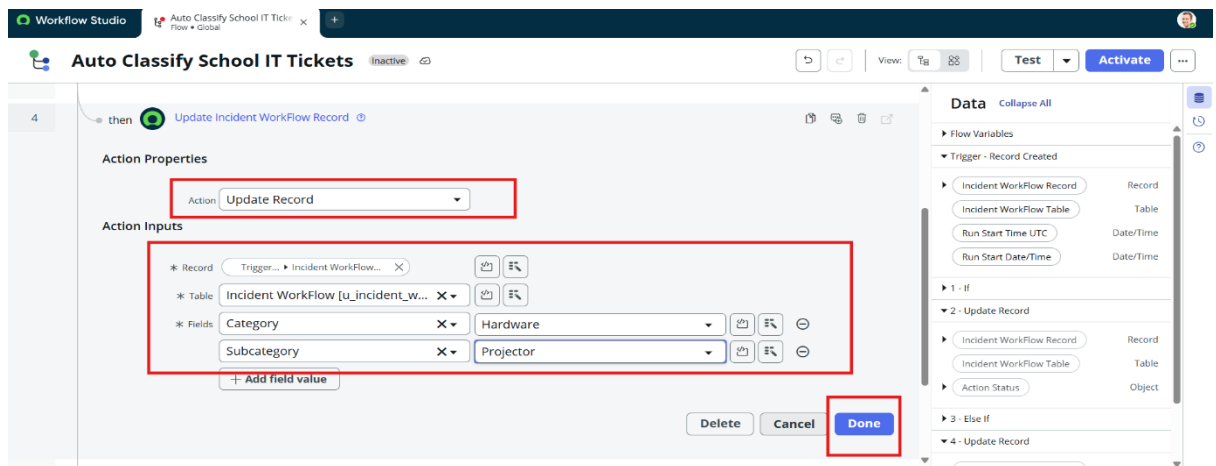


Activity : 4 Projector Issue Classification.

- Now Select the **Else If** Condition.
- Condition label : SD is **Projector or Hardware**.
- Condition : **Short description Contains Prajector**.
- Then Click the **Done** button.
- Now Action is an **Update record**.
- Set **Category = Hardware**.



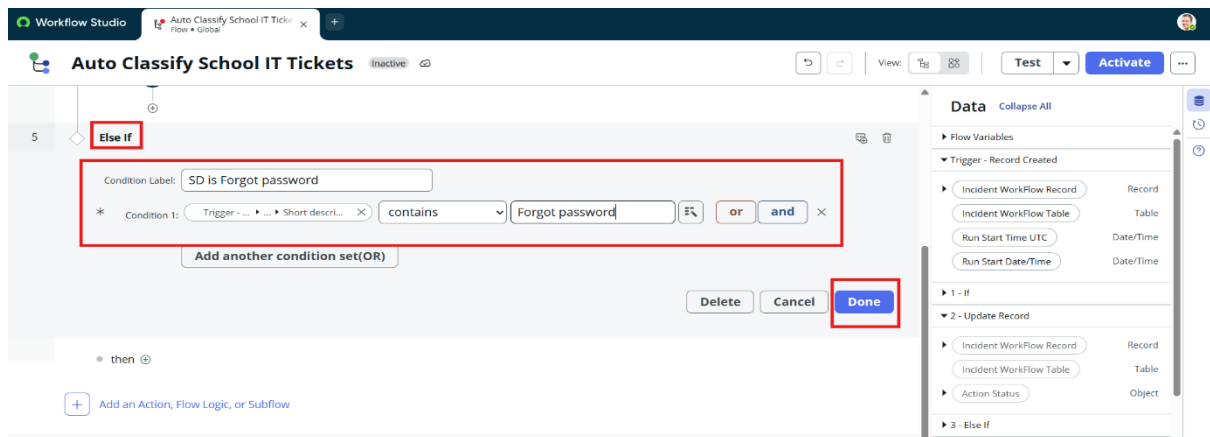
- Set Subcategory = Projector.

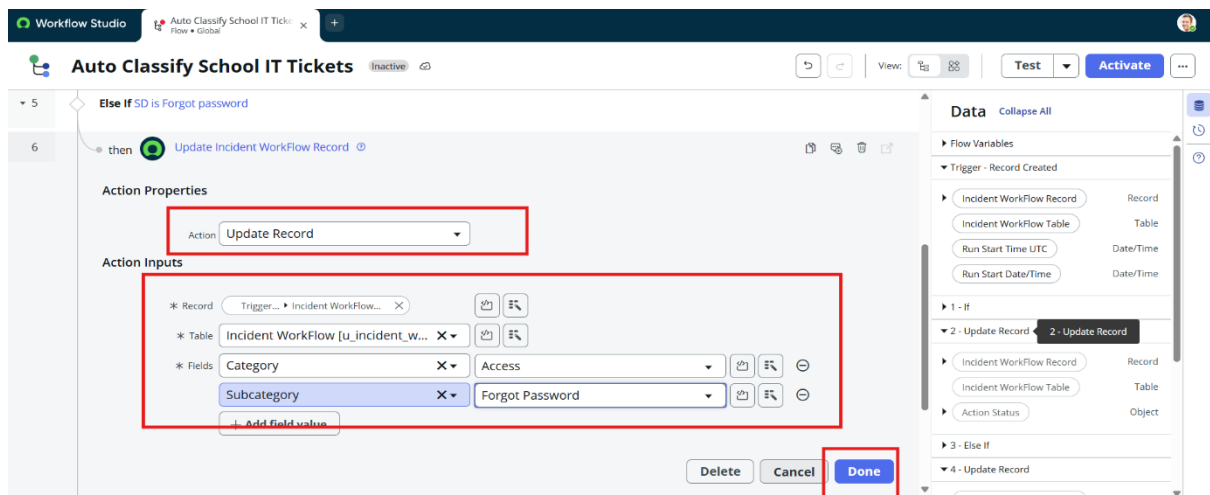


- Then Click the **Done** button.

Activity : 5 Password Issue Classification.

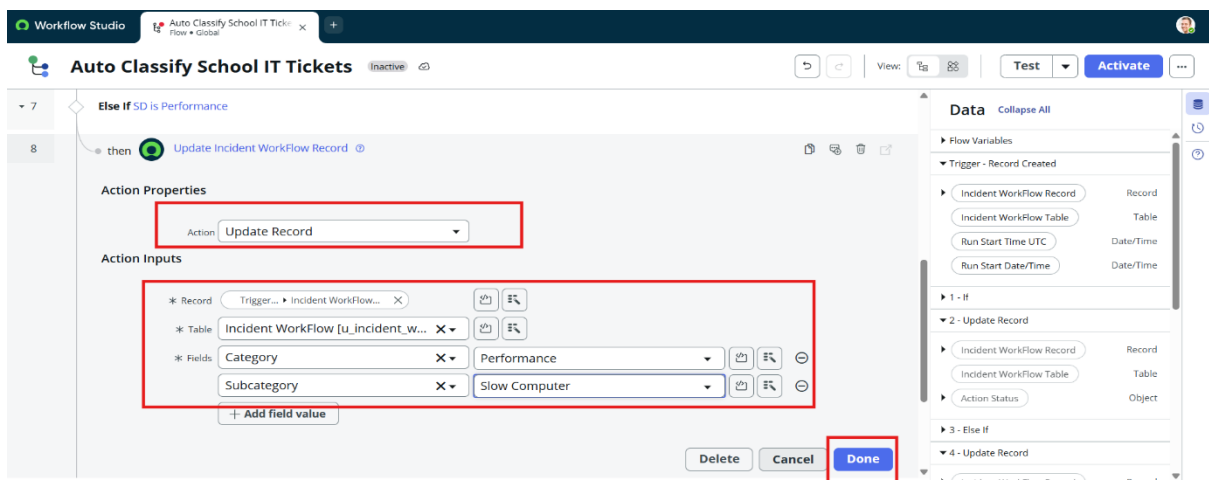
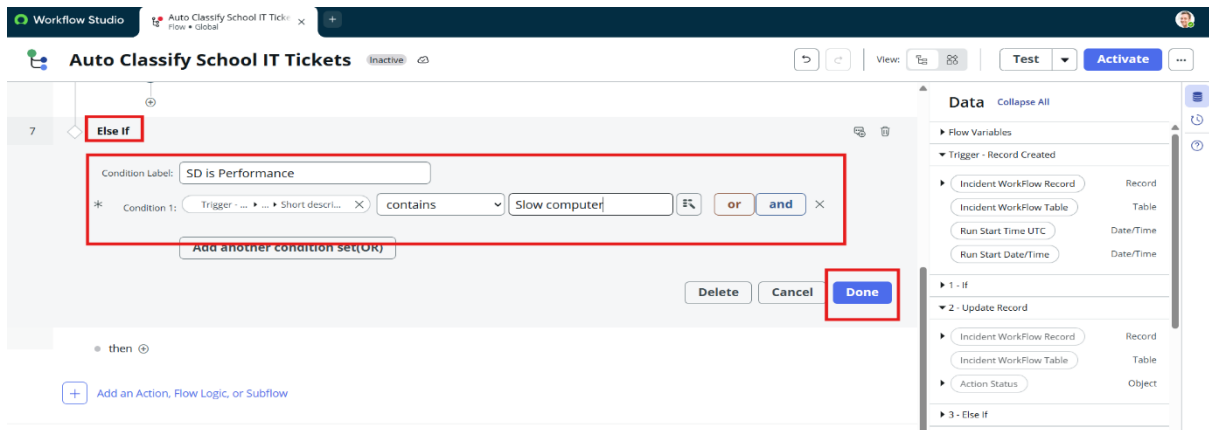
- Now again select **Else If** condition.
- Condition label : SD is **Forgot password**.
- Condition : **Short description contains Forgot password**.
- Then Click the **Done** button.
- Now Action is an **Update Record**.
- Set **Category** = **Access**.
- Set **Subcategory** = **Forgot Password**.
- Then Click the **Done** button.





Activity : 6 Slow Computer Classification.

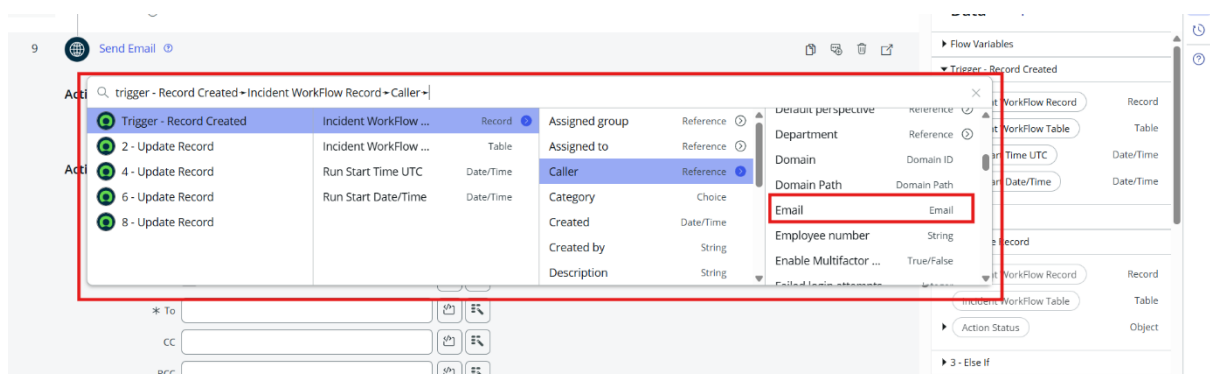
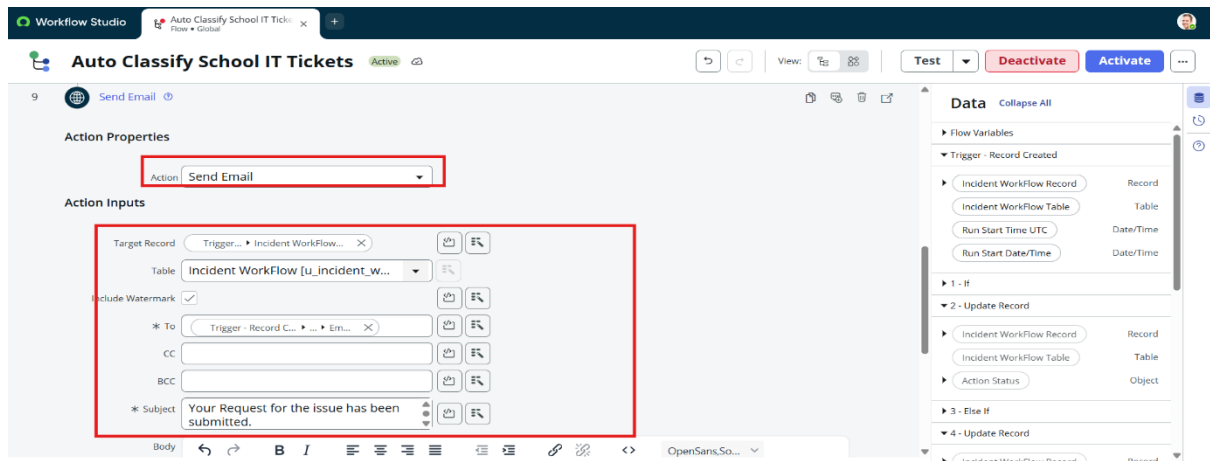
- Now Again Click the **Else If** Condition.
- Condition label : SD is **Performance**.
- Condition : **Short description contains Slow Computer**.
- Then Click the **Done** button.
- Now Action is an **Update Record**.
- Set **Category** = **Performance**.
- Set **Subcategory** = **Slow computer**.
- Then Click the **Done** button.



Activity : 7 Email Action Setup.

- After classification, an email notification is sent using Flow Designer.
- Set Action **Send Email**.
- To : **Caller → Email**.

- Subject : **Your Request for the issue has been submitted.**
- Body : **Ticket confirmation message.**
- Then Click the **Done** button.



- Now Click the **Activate** button from the top right corner.

PHASE 6 : Testing, Validation & Security.

- The objective of Phase 4 is to ensure that the developed solution works as expected, meets all functional requirements, and follows basic security and access control best practices.
- This phase validates the end-to-end automation, confirms data accuracy, and ensures that only authorized users can access or modify ticket data.

Data Validation Checks

- Mandatory fields captured correctly.
- Auto-number generated without duplication.
- Category and Subcategory values stored accurately.
- Reference fields resolve correctly to user and group tables.

Activity : 1 Testing & Validation.

Test Scenario 1

- Enter one Caller Name.
- Short Description: WiFi not working in library
- Then Save the Form.
- And Reload the Form.
- Then it fills auto as :
- Category → **Network.**

- Subcategory → **WiFi**.
- Email sent to caller Email Id.

Incident Workflow INC00511

Number: INC00511

* Caller: Alfonso Grigen

Category: Network

Subcategory: Wi-Fi

* Short description: My college Has issue with Network

Description:

State: -- None --

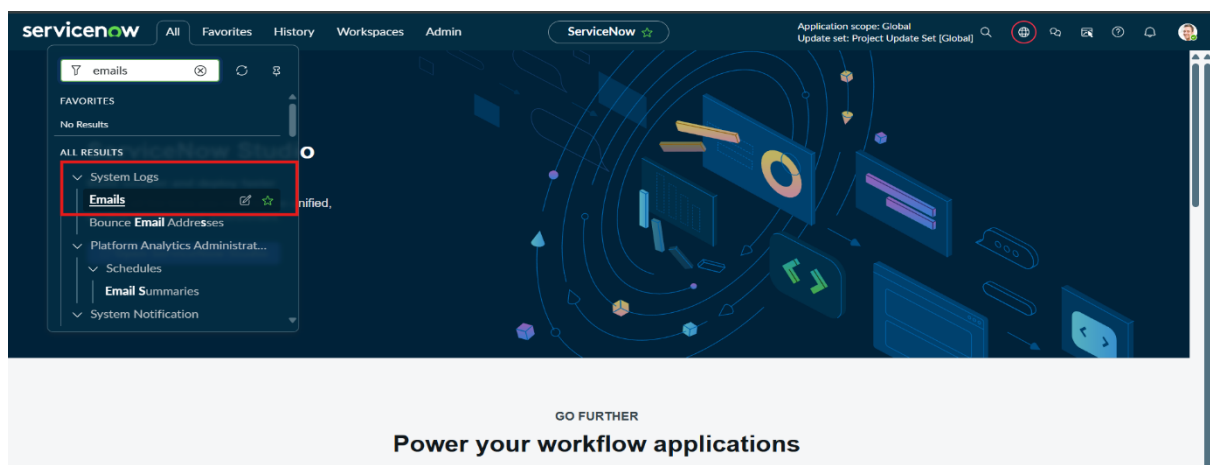
Assigned group:

Assigned to:

Update Delete

Activity : 2 Check Email Notification.

- Open All menus → Search Emails.
- Under System logs → Select Emails.
- Search with the Given Subject Name.
- Open that Record and Under Related Links Click Preview mail Option.



- Subcategory → **Projector**.
- Then Email sent to caller Mail id.

Incident Workflow INC00512

Number: INC00512

* Caller: Sam Sorokin

Category: Hardware

Subcategory: Projector

* Short description: Projector not turning on

Description:

State: -- None --

Assigned group:

Assigned to:

Update Delete

PHASE 5: Deployment & Final Documentation.

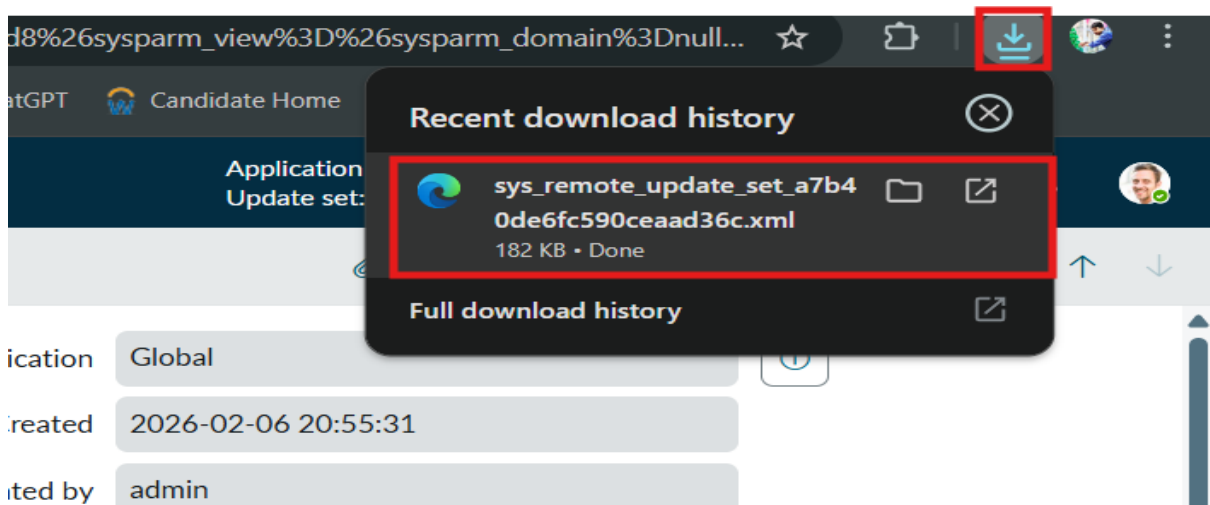
- The objective of Phase 4 is to deploy the developed solution to the target environment and finalize all project documentation.
- This phase ensures that the solution is stable, reusable, and ready for demonstration or production use, along with complete and clear documentation for future reference.

Activity : 1 Make Update set to Complete State.

- Go to All menus → Search For **Local Update Sets**.
- Under **System Update sets** Select → **Local Update sets**.

- Open the **Project Update set**.
- Make the State From **In progress** to **Complete**.
- Then **Save** the Form.
- Under Related links Click **Export to XML** Option.
- Then the Xml Format the Update set data will be **downloaded** to your system.
- We can share through WhatsApp or Mail that downloaded file.
- Then the Created Update set will transform to others in this way.

The screenshot displays the ServiceNow interface for managing update sets. The top navigation bar includes 'All', 'Favorites', 'History', 'Workspaces', 'Admin', and 'ServiceNow'. The left sidebar shows a search bar and a list of favorites, including 'System Data Management', 'Update Jobs', 'System Update Sets', 'Update Sources', 'Retrieved Update Sets', 'Update log', and 'Local Update Sets'. The main content area shows the 'Project Update Set' form. The form includes fields for 'Name' (Project Update Set), 'State' (In progress), 'Parent' (In progress), 'Release date' (Ignore), 'Install date', 'Installed from', and 'Description'. The 'State' dropdown is highlighted with a red box, showing 'In progress' as the current state and 'Complete' as an option. The 'Parent' dropdown is also highlighted with a red box, showing 'Complete' as an option. The 'Release date' dropdown is highlighted with a red box, showing 'Ignore' as an option. The 'Application' is set to 'Global'. The 'Created' date is '2026-02-06 20:55:31'. The 'Created by' is 'admin'. The 'Merged to' field is empty. The 'Update' button is visible at the bottom left.



Demo Link

<https://drive.google.com/file/d/1oIgdCxIXEiPJhyXdUJMVN2dhiNCoQHHP/view?usp=drivesdk>

Conclusion

- The **Auto Ticket Classification using Flow Designer** project successfully delivers a complete, end-to-end automation solution for a school IT helpdesk. The primary objective of reducing manual effort in ticket categorization and improving response efficiency has been fully achieved.
- Through structured requirement analysis, careful backend configuration, and no-code automation using Flow Designer, the system now automatically classifies IT tickets based on the short description provided by students. The implementation of dependent choice fields

ensures data accuracy, while automated email notifications improve communication by instantly acknowledging ticket creation to the caller.

- The solution was developed using a phased approach, ensuring clarity, stability, and ease of maintenance. Comprehensive testing and security validation confirmed that the system functions reliably, maintains data integrity, and enforces appropriate access controls.
- Overall, this project demonstrates how ServiceNow can be effectively used to build scalable, maintainable, and user-friendly automation solutions without relying on complex scripting or machine learning. The design is well-suited for small to medium environments such as educational institutions and can be easily extended in the future to include advanced features like assignment automation, SLA tracking, or predictive intelligence.